

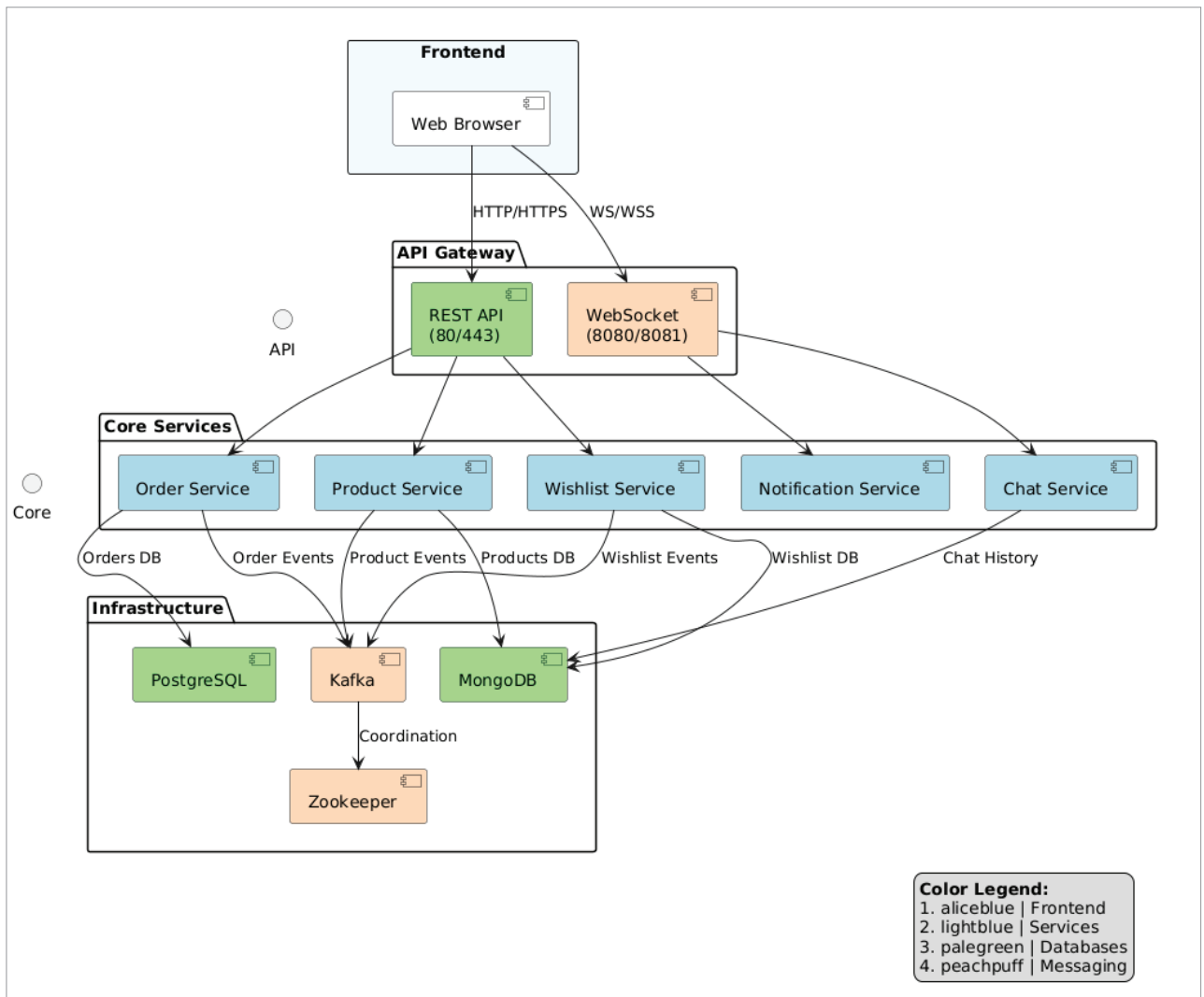


Figure 5: Frontend interaction with backend via REST and WebSockets

Zookeeper coordinates Kafka brokers. The kube-apiserver manages deployments, stateful sets, and persistent storage. Horizontal Pod Autoscaler (HPA) dynamically scales the Product service based on usage metrics.

The benefits of this deployment setup are:

- Simulates real-world DevOps pipelines using open source tools.
- Supports modular redeployment, with no need to restart the whole system.
- Simplifies scaling and fault recovery.
- Maintains clear separation between development, configuration, and runtime.

This architecture allows developers to iterate quickly, roll out features safely, and experiment with real-time load balancing and observability tools like Prometheus and Grafana.

Frontend integration and authentication

The frontend of the platform is built using Next.js, a powerful React framework that supports both server-side rendering (SSR) and static site generation (SSG). It provides fast page loads, improved SEO, and dynamic routing—key for building interactive e-commerce experiences. Figure 5 illustrates the architecture of a full-stack microservices ecosystem, where a frontend web browser communicates via an API gateway supporting both REST and WebSocket protocols.

Users can browse products, manage wishlists, place orders, and chat with sellers, all through an intuitive and responsive interface.

Frontend features

Next.js routing: Dynamic pages for products, cart, orders, chats, and login.